Kelbari

Internship Program · 2026

📈 Data Analytics internship.

A guided, portfolio-building track. Pick your tasks, build real projects, and share your work. Top performers can earn a Letter of Recommendation and a pathway to a job at Kelbari. One connected team for web development, digital marketing and graphic design, based in Bahawalpur, Pakistan.

Level 1 Level 2

Track guide & task cards
kelbari.net

Apply via the form in our
Kelbari LinkedIn post

Before you start

This checklist applies to every intern on every track. Complete it before submitting any task.

Profile & setup (do this first)

01 Complete your LinkedIn profile. If you need help, search “LinkedIn profile optimization and resume building” on YouTube and watch a tutorial before starting any project work.

02 Search “how to build a personal portfolio” on YouTube and watch a walkthrough — a simple portfolio makes your internship work easy to showcase.

03 Create your GitHub repository. Name it exactly: KELBARI (all caps). This single repository holds all your task submissions across the internship. Do not create a separate repo per task.

04 Connect your GitHub account and make sure your profile is public so mentors can review your work.

Per-task workflow (repeat for every task)

01 Step 1 — Review. Read the task card for your chosen task carefully. Understand the full feature checklist before writing a single line of code or starting a design.

02 Step 2 — Build. Develop your project, following the tech stack and requirements in the task card.

03 Step 3 — Push to GitHub. Commit your completed project to your KELBARI repository using this folder format, with no exceptions: KELBARI/[Track]-[Level or Task]-[ProjectName]/ — for example KELBARI/WebDev-L1-LandingPage/ or KELBARI/Python-Task2-BMICalculator/. Each folder must contain your source code, a README.md describing the project, and any relevant screenshots or output files.

04 Step 4 — Record a demo video. Record a short screen walkthrough of your finished project. Begin with a 2-second title card showing (1) your full name, (2) your track, and (3) the task title. After the title card, show the project working end to end — not just screenshots. For report-based tasks, narrate your document on screen.

05 Step 5 — Post to LinkedIn. Share your demo video in a LinkedIn post. Tag Kelbari and include the hashtag #kelbari, plus relevant domain hashtags (for example #webdevelopment, #datascience, #python, #cybersecurity, #internship).

06 Step 6 — Engage with peers. Watch the LinkedIn demo videos of at least two other interns from your cohort and leave a thoughtful comment on each — engage with their work, ask a question, or offer a constructive observation. Generic comments like “Great work!” don't count.

07 Step 7 — Submit. Submit your task through the submission form shared with you by Kelbari, including your GitHub repository link.

08 Step 8 — Evaluation. After the submission deadline, your work is reviewed. On successful completion you receive your Kelbari Completion Certificate and, where applicable, a Letter of Recommendation.

Key reminders

01 GitHub repo name must be exactly KELBARI (all caps) — any other name may cause your submission to be missed.

02 Follow the strict folder format KELBARI/[Track]-[Level or Task]-[ProjectName]/ — inconsistent naming delays review.

03 The first 2 seconds of every demo video must show your full name, track and task title as a static frame.

04 Thoughtful comments on at least two fellow interns' videos are required for completion.

05 Check your track's completion rule below and don't under-submit.

06 All work must be entirely your own. Plagiarism or copied work leads to cancellation of your submission and certificate.

What you receive from Kelbari

01 Guidance and mentorship from the Kelbari team throughout the program.

02 A Certificate of Completion once you finish your track's requirements.

03 For top performers: a Letter of Recommendation, a letter of appreciation, and a pathway to a job or paid role at Kelbari and related opportunities.

📈 Data Analytics

Completion rule. Complete at least 3 tasks from Level 1 or Level 2 combined. For a Letter of Recommendation, complete as many tasks as you can from both levels.

Level 1

LEVEL 1 · TASK 1

EDA on Retail Sales Data

Objective. Perform a thorough Exploratory Data Analysis on a retail sales dataset to uncover patterns, customer trends and actionable business insights.

```
Tech: Python, pandas, matplotlib, seaborn, Jupyter
```

FEATURE CHECKLIST

- [ ] Load and inspect: shape, dtypes, null check
- [ ] Descriptive statistics for all numeric columns
- [ ] Time-series: monthly and quarterly sales trends
- [ ] Customer demographics: age-group distribution, gender breakdown
- [ ] Product analysis: top 10 products; revenue by category
- [ ] Correlation heatmap
- [ ] At least one extra visualisation revealing a non-obvious insight
- [ ] Markdown observations after each chart
- [ ] Conclusion: at least 3 specific, actionable recommendations

> **Where to learn:** Search “retail sales dataset” on Kaggle (try “superstore sales”). Search “pandas EDA tutorial retail sales” and “seaborn time series plot tutorial” on YouTube.

LEVEL 1 · TASK 2

Customer Segmentation Analysis

Objective. Apply clustering to segment a customer base into distinct groups by purchasing behaviour, enabling targeted marketing.

```
Tech: Python, pandas, scikit-learn (KMeans), matplotlib, seaborn, Jupyter
```

FEATURE CHECKLIST

- [ ] Load, inspect and clean; handle missing/inconsistent data
- [ ] Descriptive statistics: average purchase value, frequency, lifetime value
- [ ] Select 2–3 behavioural features (e.g. Recency, Frequency, Monetary)
- [ ] Standardise features before clustering
- [ ] K-Means with the Elbow Method for optimal K
- [ ] Scatter plots of at least 2 feature combinations
- [ ] Profile each cluster with mean feature values
- [ ] Bar chart of customers per cluster
- [ ] Insights: a marketing action per segment

> **Where to learn:** Search “customer segmentation RFM analysis Python tutorial” on YouTube. For data, search “e-commerce customer data” on Kaggle (the “Online Retail” dataset is classic). Search “KMeans elbow method scikit-learn tutorial”.

LEVEL 1 · TASK 3

Cleaning Data

Objective. Take a deliberately messy dataset and systematically transform it into a clean, analysis-ready dataset, documenting every decision.

```
Tech: Python, pandas, numpy, Jupyter
```

FEATURE CHECKLIST

- [ ] Produce a data-quality report: nulls per column, duplicates, dtype issues, value anomalies
- [ ] Missing-data handling per column, with justification
- [ ] Duplicate removal, documenting how many were removed
- [ ] Standardise inconsistent formatting (e.g. Male/male/M → Male; dates → datetime)
- [ ] Outlier detection (IQR or Z-score); decide and document cap/remove/retain
- [ ] Correct every column's dtype
- [ ] A before-vs-after summary table
- [ ] Save the cleaned dataset to a new CSV

> **Where to learn:** Search “dirty dataset for data cleaning practice” on Kaggle (tags: data cleaning). Search “pandas data cleaning tutorial 2026” on YouTube.

LEVEL 1 · TASK 4

Sentiment Analysis

Objective. Build a model that classifies the sentiment of text (positive, negative or neutral) for insight into public opinion or customer feedback.

```
Tech: Python, pandas, scikit-learn, NLTK or TextBlob, matplotlib/seaborn, Jupyter
```

FEATURE CHECKLIST

- [ ] Load and inspect class distribution
- [ ] Text preprocessing: lowercase, punctuation/stopword removal, tokenisation, optional stemming
- [ ] TF-IDF feature extraction (explain its purpose)
- [ ] Train/test split (80/20)
- [ ] Train at least 2 classifiers: Naive Bayes + one of SVM/Logistic Regression
- [ ] Evaluate: accuracy, precision, recall, F1, confusion matrix per model
- [ ] Bar chart of sentiment distribution; WordCloud per class
- [ ] Error analysis: 5 misclassified examples and why
- [ ] Conclusion: best model and a real-world application

> **Where to learn:** Search “sentiment analysis dataset” on Kaggle (Twitter, Amazon reviews, IMDB). Search “Python sentiment analysis NLP tutorial scikit-learn” on YouTube. Reference the NLTK and TextBlob docs.

Level 2

LEVEL 2 · TASK 1

Predicting House Prices with Linear Regression

Objective. Build and evaluate a linear regression model that predicts house prices from area, location, rooms and age — end to end.

`Tech: Python, pandas, scikit-learn, matplotlib, seaborn, Jupyter`

FEATURE CHECKLIST

- [ ] Load and EDA: null check, descriptive statistics, target distribution
- [ ] Feature-selection discussion with reasoning
- [ ] Handle missing values; one-hot encode categoricals
- [ ] Correlation heatmap vs price
- [ ] Train/test split (80/20)
- [ ] Train a Linear Regression model
- [ ] Evaluate: MSE, RMSE, R^2
- [ ] Scatter plot of actual vs predicted
- [ ] Residual plot
- [ ] Coefficient analysis: highest positive/negative impact
- [ ] (Bonus) Compare against Ridge or Lasso

> **Where to learn:** Search “house price prediction dataset” on Kaggle (Ames Housing is excellent). Search “linear regression house price prediction Python tutorial” on YouTube. Reference scikit-learn for LinearRegression and the metrics.

LEVEL 2 · TASK 2

Wine Quality Prediction

Objective. Train and compare classifiers to predict a wine quality score from physicochemical properties like acidity, density and alcohol.

```
Tech: Python, pandas, numpy, scikit-learn, seaborn, matplotlib, Jupyter
```

FEATURE CHECKLIST

- [ ] Load and inspect; check quality-score distribution
- [ ] EDA: feature distributions and a correlation heatmap
- [ ] Discuss class imbalance
- [ ] Feature engineering: bin scores into binary or 3-class, with justification
- [ ] Stratified train/test split
- [ ] Train 3 classifiers: Random Forest, SGD and SVC
- [ ] Evaluate each: accuracy, classification report, confusion matrix
- [ ] Feature-importance chart for Random Forest
- [ ] A side-by-side comparison table
- [ ] Conclusion: most suitable model and why

> **Where to learn:** Search “Wine Quality dataset UCI” (on Kaggle and the UCI repository). Search “wine quality prediction Python Random Forest SVM tutorial” on YouTube.

LEVEL 2 · TASK 3

Fraud Detection

Objective. Build a pipeline to detect fraudulent transactions from a heavily imbalanced dataset, treating class imbalance as the core challenge.

```
Tech: Python, pandas, scikit-learn, imbalanced-learn (SMOTE), matplotlib, seaborn, Jupyter
```

FEATURE CHECKLIST

- [ ] Analyse class imbalance (what percentage is fraud?)
- [ ] EDA: amount distribution for fraud vs non-fraud; time-of-day analysis
- [ ] Explain why plain accuracy misleads on imbalanced data
- [ ] Apply an imbalance technique: SMOTE, undersampling or class_weight='balanced'
- [ ] Stratified train/test split
- [ ] Train at least 2 models: Logistic Regression + one of Decision Tree/Random Forest/XGBoost
- [ ] Evaluate: precision, recall, F1 and AUC-ROC
- [ ] Explain which metric matters most (recall vs precision)
- [ ] Feature-importance or coefficient analysis
- [ ] Discuss scalability to high transaction volumes

> **Where to learn:** Search “credit card fraud detection dataset” on Kaggle (a 284,807-row benchmark). Search “fraud detection Python imbalanced dataset SMOTE tutorial” on YouTube. Reference the imbalanced-learn docs.

LEVEL 2 · TASK 4

App Market Analysis (Play Store)

Objective. Analyse the app-store ecosystem — cleaning messy real-world data, exploring categories, analysing ratings and pricing, and running sentiment analysis on reviews.

```
Tech: Python, pandas, numpy, matplotlib, seaborn, TextBlob or VADER, Jupyter
```

FEATURE CHECKLIST

- [ ] Load the apps dataset and the reviews dataset separately
- [ ] Clean: fix types (e.g. “Installs” as “10,000+” strings), handle nulls, remove duplicates
- [ ] Category analysis: distribution across categories; most saturated?
- [ ] Ratings analysis: rating distribution; average by category
- [ ] Size vs installs scatter plot
- [ ] Pricing: paid vs free; price distribution; rough revenue by category
- [ ] Sentiment analysis on reviews (TextBlob or VADER)
- [ ] Sentiment by category
- [ ] At least one plotly interactive chart (recommended)
- [ ] Conclusion: 3 insights for a developer launching a new app

> **Where to learn:** Search “Google Play Store dataset” on Kaggle. Search “Google Play Store EDA Python tutorial” on YouTube. For sentiment, search “VADER sentiment analysis Python tutorial”.

LEVEL 2 · TASK 5

Autocomplete & Autocorrect Analytics

Objective. Analyse the efficiency and accuracy of autocomplete and autocorrect using NLP, implementing and comparing multiple approaches on a real text dataset.

```
Tech: Python, pandas, NLTK, pyspellchecker or textdistance, collections, matplotlib, Jupyter
```

FEATURE CHECKLIST

- [ ] Collect or download a large text corpus
- [ ] NLP preprocessing: tokenisation, lowercasing, punctuation/stopword removal
- [ ] Autocomplete: a frequency-based n-gram model that predicts the next word
- [ ] Test autocomplete on 10+ prefixes; show top 3 predictions each
- [ ] Autocorrect: edit-distance correction via pyspellchecker or custom Levenshtein
- [ ] Test autocorrect on 20+ misspelled words; measure accuracy
- [ ] Define and calculate precision and recall for both
- [ ] Compare at least 2 approaches
- [ ] Bar chart of top 20 words; a before/after confusion matrix
- [ ] Discuss limitations vs production systems

> **Where to learn:** For a corpus, use Project Gutenberg (gutenberg.org) or a Kaggle text corpus. Search “Python autocomplete n-gram model tutorial” and “Python spell checker pyspellchecker tutorial”. Reference the NLTK docs for ngrams and FreqDist.

Contact & support

Questions during your internship? Reach the Kelbari team here.

Email	kelbariofficial@gmail.com
Support	support.kelbari@gmail.com
WhatsApp	+92 301 9233246 (available 24/7)
Website	kelbari.net
LinkedIn	Follow the Kelbari company page on LinkedIn

The purpose of this internship is to learn and grow. Treat every task with professional care, ask for guidance when you need it, and use the work to strengthen your portfolio and confidence. **— The Kelbari Team**